

PD structures

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p -adic cohomology theory

$$X/\mathbb{F}_p, \quad \tilde{X}/\mathbb{Z}_p, \quad H^i(X) = H_{\text{dR}}^i(\tilde{X}) \quad ?$$

S PD scheme, $\text{Cris}(X/S)$

§1. Divided powers

Def. I ideal of A . $\gamma_i: I \rightarrow A$ is a PD structure on I if

i) $\gamma_0(x) = 1, \gamma_1(x) = x, \gamma_i(x) \in I, \forall i \geq 1$

ii) $\gamma_n(x+y) = \sum_{i+j=n} \gamma_i(x) \gamma_j(y)$

iii) $\gamma_n(rx) = r^n \gamma_n(x), r \in A$

iv) $\gamma_i(x) \gamma_j(x) = \frac{(i+j)!}{i!j!} \gamma_{i+j}(x)$

v) $\gamma_p(\gamma_q(x)) = \frac{(pq)!}{p!(q!)^p} \gamma_{pq}(x)$

(I, γ) PD ideal; (A, I, γ) PD ring

Def. $\psi: (A, I, \gamma) \rightarrow (B, J, \delta)$ PD mor. if

- $\psi(I) \subset J$

- $\psi \circ \gamma = \delta \circ \psi$

Def. (S, I, γ) PD scheme.

Rem - If $0 \subset A, \exists!$ PD str. on any I .

- If $\mathbb{Z} \subset A$, then I has a PD str. iff $\forall x \in I, \frac{x^k}{k!} \in I, \forall k$.
 $\forall n, A[n] = 0$

- If A is a DVR of mixed $\text{char}^{(0,p)}$ w/ uniformizer π , $p = u\pi^e$

(π) has a PD str. iff $e \leq p-1$.

- $M=0$ in A , (I, γ) PD, $m! \gamma_m(x) = x^m = 0$.

Def. (I, γ) PD, $J \subset I$, J is a sub PD ideal if $\forall i \geq 1, \gamma_i(J) \subset J$.

Prop. If M is an A -module, $\exists (\Gamma(M), \Gamma'(M), \delta)$ which is an A -PD algebra, s.t.

$\forall (B, J, \delta)$ another A -PD-alg, $\psi: M \rightarrow \Gamma'(M)$, $M \xrightarrow{\bar{\psi}} J$ linear

$\exists!$ PD mor. $\Gamma(M) \xrightarrow{\psi} B$ s.t. $\psi \circ \psi = \bar{\psi}$.

§1. Divided powers

Prop. (A, I, γ) PD ring, $J \subset A$ ideal. Suppose $I \cap J \subset I$ sub PD ideal.

Then $\exists!$ PD str. $\bar{\gamma}$ on $\bar{I} = I(A/J)$ s.t. $(A, I, \gamma) \rightarrow (A/J, \bar{I}, \bar{\gamma})$ is a PD map.

Prop. $(A, I, \gamma), (A, J, \delta)$ two PD structures, $\gamma|_{I \cap J} = \delta|_{I \cap J}$, then $\exists!$ PD str.

on $I+J$ inducing γ, δ .

Pr. $0 \rightarrow I \cap J \rightarrow I \oplus J \rightarrow I+J \rightarrow 0$

$x \in I, g(x) = \sum \gamma_i(x) T^i \in A[[T]]$

$y \in J, h(y) = \sum \delta_j(y) T^j \in A[[T]]$

$g(x+x') = g(x)g(x')$

if $f \in \exp(A)$, $(rf)(T) = f(rT)$, $I \oplus J \xrightarrow{(g,h)} \exp(A)$

$\sim e: I+J \rightarrow \exp(A)$, $\gamma \in I+J, e(\gamma) = \sum (\xi_n(\gamma)) T^n$.

Def (A, I, γ) be PD ring, B an A -alg. γ extends to B if $\exists$ PD str.

$\bar{\gamma}$ on IB making $(A, I, \gamma) \rightarrow (B, IB, \bar{\gamma})$ a PD map.

Rem $J \subset A$, $I \cap J \subset I$ NOT a sub PD ideal, then γ DOESN'T EXTEND to A/J .

Def/Prop (A, I, γ) , B as before. (J, δ) a PD ideal of B . TFAE.

1) γ extends to B & $\bar{\gamma} = \delta$ on $IB \cap J$.

2) $K = IB + J$ has PD str. $\bar{\delta}$ s.t.

$$(A, I, \gamma) \rightarrow (B, K, \bar{\delta}) \leftarrow (B, J, \delta)$$

3) 2 But for $K \supset IB + J$.

In this case we say γ and δ are compatible.



§2. Crystalline site & topos.

Fix p prime. All schemes are killed by a power of p .

Let (S, I, γ) be a fixed PD scheme.

eg. $S = W_n(k)$, $I = (p)$, γ .

Def Let X be an S -scheme s.t. γ extends to X .

Cris (X/S) : objects. $\left(\begin{array}{c} U \xrightarrow{\text{cl.}} T, \delta \\ \text{open} \downarrow \\ X \end{array} \right) : U \subset X \text{ Zariski open.}$
 δ is a PD str. on $I U \rightarrow T$
 compatible with γ .

Morphism $(u \hookrightarrow T, \delta) \longrightarrow (u' \hookrightarrow T', \delta')$

$$\begin{array}{ccc} \text{or } u & \hookrightarrow & T \\ x \downarrow & \text{PD map} & \downarrow \\ u' & \hookrightarrow & T' \end{array}$$

Coverings $\{u_i: T_i \rightarrow T\}$ u_i mor. in $\text{Cris}(X/S)$

$$T_i \xrightarrow[\text{open}]{} T, \quad T = \bigcup T_i$$

Def. $(X/S)_{\text{cris}} = \text{Sh}(\text{Cris}(X/S)).$

Prop. $F \in (X/S)_{\text{cris}} \iff \forall (u, T, \delta) \in \text{Cris}(X/S) \quad F_T \in \text{Sh}(T_{\text{Zar}})$

$$\forall u: T_i \rightarrow T \text{ in } \text{Cris}(X/S), \quad u^{-1}F_T \rightarrow T_i \text{ s.t.}$$

$$i) \quad v^{-1}u^{-1}F_T \rightarrow v^{-1}F_{T_1} \rightarrow F_{T_2}$$

$$\begin{array}{ccc} \downarrow & \text{PD map} & \downarrow \\ (uv)^{-1}F_T & \longrightarrow & F_{T_2} \end{array}$$

$$ii) \quad \text{if } T_i \xrightarrow[u]{\text{open}} T, \quad u^{-1}F_T \xrightarrow{\sim} F_{T_i}.$$

Cor. $(X/S)_{\text{cris}}$ "has enough points"

$F \rightarrow g$ is an isom. iff $\forall x \in X, \exists x \in U \subset X, (u \hookrightarrow T, \delta) \in \text{Cris}(X/S)$

$$(F_T)_x \xrightarrow{\sim} (g_T)_x.$$

Ex. 1. $T \mapsto \mathcal{O}_T$ is a sheaf $\mathcal{O}_{X/S}$

2. $T \mapsto \mathcal{O}_U$ is a sheaf $\mathcal{O}_X$.

3. $T \mapsto \ker(\mathcal{O}_T \rightarrow \mathcal{O}_U) = I_U \hookrightarrow T$, $\mathcal{I}_{X/S} \subset \mathcal{O}_{X/S}$.

